

Linear Bandits

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Outline

1 Recap: Linear Bandits

- Setting
- LinUCB
- An Optimal Regret Bound

2 Analysis

- Regret Analysis
- Confidence Analysis

3 Deep Learning Connection: NTK Bandits

Recap: The “Assumption Ladder” for Linear RL

- **Linear Feature Assumptions:** We explored a hierarchy of conditions for RL:
 - Bellman completeness
 - All-policy realizability
 - Linear Q^* realizability
- **The Goal:** Can we achieve sample complexities that are $\text{poly}(d, H)$?
Crucially, we sought no dependence on the sizes of the state/action spaces ($|S|$ or $|A|$).
- **Today:** We strip away the horizon complexity and focus on the $H = 1$ case.
 - **Linear Bandits:** How do we explore efficiently in large/infinite action spaces?

Handling Large Action Spaces

- On each round, we must **choose a decision** $x_t \in D \subset \mathbb{R}^d$.
- Obtain a reward $r_t \in [-1, 1]$, where

$$r_t = \mu^\star \cdot x_t + \eta_t$$

with i.i.d noise η_t .

- the conditional expectation of r_t is linear,

$$\mathbb{E}[r_t | x_t = x] = \mu^\star \cdot x$$

Our Objective

If $x_0, \dots, x_{T-1}$ are our decisions, then our **cumulative regret** is

$$R_T = T \mu^* \cdot x^* - \sum_{t=0}^{T-1} \mu^* \cdot x_t$$

where $x^* \in D$ is an optimal decision for μ^* , i.e.

$$x^* \in \operatorname{argmax}_{x \in D} \mu^* \cdot x$$

The “Confidence Ball”

After t rounds, define our uncertainty region BALL_t : with center, $\hat{\mu}_t$, and shape, Σ_t , using the λ -regularized least squares solution:

$$\hat{\mu}_t = \underset{\mu}{\operatorname{argmin}} \sum_{\tau=0}^{t-1} (\mu \cdot x_\tau - r_\tau)^2 + \lambda \|\mu\|_2^2$$

$$= \Sigma_t^{-1} \sum_{\tau=0}^{t-1} r_\tau x_\tau,$$

$$\Sigma_t = \lambda I + \sum_{\tau=0}^{t-1} x_\tau x_\tau^\top, \text{ with } \Sigma_0 = \lambda I.$$

$$\{(x_\tau, r_\tau)\}_{\tau=0}^{t-1}$$

$$\bar{\Sigma}_t = \frac{1}{t} \Sigma_t$$

$$\|\hat{\mu}_t - \mu^*\|_{\bar{\Sigma}_t}^2 \leq \frac{d\sigma^2}{t}$$

$$\|\hat{\mu}_t - \mu^*\|_{\Sigma_t}^2 \leq \beta_t$$

Define the uncertainty region:

$$\text{BALL}_t = \left\{ \mu \mid (\hat{\mu}_t - \mu)^\top \Sigma_t (\hat{\mu}_t - \mu) \leq \beta_t \right\},$$

where β_t is a parameter of the algorithm.

LinUCB (the algo)

- ① Input: λ, β_t
- ② Initialize: $\Sigma_0 = \lambda I, b_0 = 0$
- ③ For $t = 0, 1, \dots$
 - ① Compute $\hat{\mu}_t = \Sigma_t^{-1} b_t$
 - ② Execute

$$x_t = \operatorname{argmax}_{x \in D} \max_{\mu \in \text{BALL}_t} \mu \cdot x$$

and observe the reward r_t . Equivalently,

$$x_t \in \operatorname{argmax}_{x \in D} \left(\hat{\mu}_t \cdot x + \sqrt{\beta_t} \sqrt{x^\top \Sigma_t^{-1} x} \right)$$

- ③ Update: $\Sigma_{t+1} = \Sigma_t + x_t x_t^\top$ and $b_{t+1} = b_t + r_t x_t$.

LinUCB Regret Bound

Sublinear regret: $R_T \leq O^*(d\sqrt{T})$

poly dependence on d , no dependence on the cardinality $|D|$.

$\sqrt{KT}$ for Karmaris

Theorem

Suppose: $|\mu^* \cdot x| \leq 1$ and $\|x\| \leq B$ for all $x \in D$; that the noise is σ^2 sub-Gaussian; and that $\|\mu^*\| \leq W$. Set $\lambda = \sigma^2/W^2$ and

$\beta_t \approx O(d \log t)$

$$\beta_t := c \left(\sigma^2 d \log \left(1 + \frac{tB^2W^2}{d\sigma^2} \right) + \sigma^2 \log(1/\delta) \right)$$

With probability greater than $1 - \delta$, for all $T \geq 0$,

$\frac{R_T}{T} \approx \frac{O(d\sqrt{T})}{T}$

$$R_T \leq c' \sigma \sqrt{T} \left(d \log \left(1 + \frac{TB^2W^2}{d\sigma^2} \right) + \log(4/\delta) \right)$$

where c, c' are absolute constants.

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In establishing the upper bounds there are two main propositions from which the upper bounds follow. The first is in showing that the confidence region is valid.

Proposition

(Confidence) Let $\delta > 0$. We have that for our choice of β_t ,

$$\Pr(\forall t, \mu^* \in \text{BALL}_t) \geq 1 - \delta.$$

Equivalently, for all $t \geq 0$:

$$(\hat{\mu}_t - \mu^*)^\top \Sigma_t (\hat{\mu}_t - \mu^*) \leq \beta_t$$

Sum of Squares Regret Bound

Assuming the confidence event holds, the following controls on the growth of the regret.

Proposition

(Sum of Squares Regret Bound) Define:

$$\text{regret}_t = \mu^* \cdot x^* - \mu^* \cdot x_t$$

Suppose $\|x\| \leq B$ for $x \in D$. Suppose β_t is increasing and larger than 1. Suppose $\mu^ \in \text{BALL}_t$ for all t , then*

$$\sum_{t=0}^{T-1} \text{regret}_t^2 \leq 4\beta_T d \log \left(1 + \frac{TB^2}{d\lambda} \right)$$

$$\approx O(d \log T)^2$$

Completing the Proof

$$a \cdot b \leq \|a\| \|b\|$$

Proof of Theorem 1: With the two previous Propositions, along with the Cauchy-Schwarz inequality, we have, with probability at least $1 - \delta$,

$$R_T = \sum_{t=0}^{T-1} \text{regret}_t \leq \sqrt{T \sum_{t=0}^{T-1} \text{regret}_t^2} \leq \sqrt{4T\beta_T d \log \left(1 + \frac{TB^2}{d\lambda}\right)}.$$

The remainder of the proof follows from using our chosen value of β_T and algebraic manipulations.

$$O(d \log T)^2 \quad d \sqrt{T}$$

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“Width” of Confidence Ball

Lemma

If $\mu \in \text{BALL}_t$ and $x \in D$, then

$$|(\mu - \hat{\mu}_t)^\top x| \leq \sqrt{\beta_t x^\top \Sigma_t^{-1} x}$$

$$|(\mu - \hat{\mu}_t)^\top x| = |(\mu - \hat{\mu}_t) \Sigma^{\frac{1}{2}} \cdot (\Sigma^{-\frac{1}{2}} x)|$$

$$\leq \|\Sigma^{\frac{1}{2}} (\mu - \hat{\mu}_t)\|_2 \cdot \|\Sigma^{-\frac{1}{2}} x\|_2$$

$$\leq \beta_t \sqrt{x^\top \Sigma_t^{-1} x}$$

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$$|(\mu - \hat{\mu}_t)^\top x| \leq \sqrt{\beta_t x^\top \Sigma_t^{-1} x}$$

Proof: By Cauchy-Schwarz, we have:

$$\begin{aligned} |(\mu - \hat{\mu}_t)^\top x| &= |(\mu - \hat{\mu}_t)^\top \Sigma_t^{1/2} \Sigma_t^{-1/2} x| \\ &\leq \|\Sigma_t^{1/2}(\mu - \hat{\mu}_t)\| \|\Sigma_t^{-1/2} x\| = \|\Sigma_t^{1/2}(\mu - \hat{\mu}_t)\| \sqrt{x^\top \Sigma_t^{-1} x} \leq \sqrt{\beta_t x^\top \Sigma_t^{-1} x} \end{aligned}$$

where the last inequality holds since $\mu \in \text{BALL}_t$.

Instantaneous Regret Lemma

Normalized width at time t for decision x_t :

$$w_t := \sqrt{x_t^\top \Sigma_t^{-1} x_t}$$

Lemma

Fix $t \leq T$. If $\mu^* \in \text{BALL}_t$, then

$$\text{regret}_t \leq 2\sqrt{\beta_t} w_t$$

$$\text{regret}_t \leq 2 \min(\sqrt{\beta_t} w_t, 1) \leq 2\sqrt{\beta_T} \min(w_t, 1)$$

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Proof: Let $\tilde{\mu} \in \arg\max_{\mu \in \text{BALL}_t} \mu^\top x_t$. By the choice of x_t and the assumption $\mu^* \in \text{BALL}_t$, we have **optimism**:

$$\tilde{\mu}^\top x_t = \max_{x \in D} \max_{\mu \in \text{BALL}_t} \mu^\top x \geq (\mu^*)^\top x^*$$

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We can decompose the regret by adding and subtracting $\hat{\mu}_t$:

$$\begin{aligned} \text{regret}_t &= (\mu^*)^\top x^* - (\mu^*)^\top x_t \leq (\tilde{\mu} - \mu^*)^\top x_t \\ &= (\tilde{\mu} - \hat{\mu}_t)^\top x_t + (\hat{\mu}_t - \mu^*)^\top x_t \leq 2\sqrt{\beta_t} w_t \end{aligned}$$

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Note: Since rewards $r_t \in [-1, 1]$, regret_t is trivially bounded by 2, yielding the “min”.

Geometric Argument: Part 1

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$$\Sigma_0 := \lambda I$$

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Lemma

We have:

$$\det \Sigma_T = \det \Sigma_0 \prod_{t=0}^{T-1} (1 + w_t^2).$$

$$w_t = \sqrt{x_t^T (\Sigma_t^{-1}) x_t}$$
$$\Sigma_{t+1} = \Sigma_t + x_t x_t^T$$

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Lemma

We have:

$$\det \Sigma_T = \det \Sigma_0 \prod_{t=0}^{T-1} (1 + w_t^2).$$

Proof: By the definition of Σ_{t+1} , we have

$$\begin{aligned} \det \Sigma_{t+1} &= \det(\Sigma_t + x_t x_t^\top) = \det(\Sigma_t^{1/2} (I + \Sigma_t^{-1/2} x_t x_t^\top \Sigma_t^{-1/2}) \Sigma_t^{1/2}) \\ &= \det(\Sigma_t) \det(I + \Sigma_t^{-1/2} x_t (\Sigma_t^{-1/2} x_t)^\top) = \det(\Sigma_t) \det(I + v_t v_t^\top), \\ &= \det(\Sigma_t) (1 + \|v_t\|^2) \end{aligned}$$

where $v_t := \Sigma_t^{-1/2} x_t$. Now observe that $v_t^\top v_t = w_t^2$.

Geometric Argument: Part 2

Lemma

For any sequence where $\|x_t\|_2 \leq B$, we have:

$$\log \left(\frac{\det \Sigma_{T-1}}{\det \Sigma_0} \right) = \log \det \left(I + \frac{1}{\lambda} \sum_{t=0}^{T-1} x_t x_t^\top \right) \leq d \log \left(1 + \frac{TB^2}{d\lambda} \right)$$

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Proof: Let $\sigma_1, \dots, \sigma_d$ be the eigenvalues of $\sum_{t=0}^{T-1} x_t x_t^\top$. We have:

$$\sum_{i=1}^d \sigma_i = \text{Tr} \left(\sum_{t=0}^{T-1} x_t x_t^\top \right) = \sum_{t=0}^{T-1} \|x_t\|_2^2 \leq TB^2$$

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Handwritten notes: $(\sigma_1 \dots \sigma_d)^{1/d} \leq \frac{1}{d} \sum \sigma_i$
 $\frac{1}{d} \sum \sigma_i \leq \log \sigma_i \leq \log \left(\frac{1}{d} \sum \sigma_i \right)$

Next, apply the AM-GM inequality to the log-determinant:

$$\log \det \left(I + \frac{1}{\lambda} \sum_{t=0}^{T-1} x_t x_t^\top \right) = \sum_{i=1}^d \log \left(1 + \frac{\sigma_i}{\lambda} \right) \leq d \log \left(1 + \frac{1}{d} \sum_{i=1}^d \frac{\sigma_i}{\lambda} \right) \leq d \log \left(1 + \frac{TB^2}{d\lambda} \right)$$

Proving “Sum of Squares Regret”

$$r_t \leq 2\sqrt{\beta_t} \min(w_t, 1)$$

Proof of Proposition 3:

Assume $\mu^* \in \text{BALL}_t$ for all t . We have:

$$\sum_{t=0}^{T-1} \text{regret}_t^2 \leq \sum_{t=0}^{T-1} 4\beta_t \min(w_t^2, 1) \leq 4\beta_T \sum_{t=0}^{T-1} \min(w_t^2, 1)$$

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Proof of Proposition 3:

Assume $\mu^* \in \text{BALL}_t$ for all t . We have:

for $\alpha \leq 1$

$\alpha \leq 2 \lg(1/\epsilon)$

$$\begin{aligned} \sum_{t=0}^{T-1} \text{regret}_t^2 &\leq \sum_{t=0}^{T-1} 4\beta_t \min(w_t^2, 1) \leq 4\beta_T \sum_{t=0}^{T-1} \min(w_t^2, 1) \\ &\leq 8\beta_T \sum_{t=0}^{T-1} \ln(1 + w_t^2) \leq 8\beta_T \log \left(\frac{\det \Sigma_{T-1}}{\det \Sigma_0} \right) \end{aligned}$$

||
log $\frac{\det \Sigma_{T-1}}{\det \Sigma_0}$

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where the first inequality follows from Lemma 5; the second from that β_t is an increasing function of t ; the third uses that for $0 \leq y \leq 1$, $\ln(1 + y) \geq y/2$; and the final two follow from Lemmas 6 and 7.

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Confidence [Proof of Proposition 2]

Since $r_\tau = x_\tau \cdot \mu^* + \eta_\tau$, we decompose the error:

$$\hat{\mu}_t - \mu^* = \Sigma_t^{-1} \sum_{\tau=0}^{t-1} x_\tau (x_\tau^\top \mu^* + \eta_\tau) - \mu^*$$

$$= \Sigma_t^{-1} (\Sigma_t - \lambda I) \mu^* - \mu^* + \Sigma_t^{-1} \sum_{\tau=0}^{t-1} \eta_\tau x_\tau$$

$$= -\lambda \Sigma_t^{-1} \mu^* + \Sigma_t^{-1} \sum_{\tau=0}^{t-1} \eta_\tau x_\tau$$

$$\Sigma_t = \left(\sum_{\tau=0}^{t-1} x_\tau x_\tau^\top \right) + \lambda I$$

Confidence [Proof of Proposition 2]

Since $r_\tau = x_\tau \cdot \mu^\star + \eta_\tau$, we decompose the error:

$$\begin{aligned}\hat{\mu}_t - \mu^\star &= \Sigma_t^{-1} \sum_{\tau=0}^{t-1} x_\tau (x_\tau^\top \mu^\star + \eta_\tau) - \mu^\star \\ &= \Sigma_t^{-1} (\Sigma_t - \lambda I) \mu^\star - \mu^\star + \Sigma_t^{-1} \sum_{\tau=0}^{t-1} \eta_\tau x_\tau \\ &= -\lambda \Sigma_t^{-1} \mu^\star + \Sigma_t^{-1} \sum_{\tau=0}^{t-1} \eta_\tau x_\tau\end{aligned}$$

Using the norm $\|z\|_{\Sigma_t} := \sqrt{z^\top \Sigma_t z}$, the triangle inequality gives:

$$\begin{aligned}\|\hat{\mu}_t - \mu^\star\|_{\Sigma_t} &\leq \|\lambda \Sigma_t^{-1} \mu^\star\|_{\Sigma_t} + \left\| \Sigma_t^{-1} \sum_{\tau=0}^{t-1} \eta_\tau x_\tau \right\|_{\Sigma_t} \\ &\leq \sqrt{\lambda} \|\mu^\star\|_2 + \underbrace{\left\| \sum_{\tau=0}^{t-1} \eta_\tau x_\tau \right\|_{\Sigma_t^{-1}}}_{??}\end{aligned}$$

Self-Normalizing Sum

Lemma (Self-Normalized Bound for Vector-Valued Martingales)

Suppose $\{\varepsilon_i\}_{i=1}^\infty$ is a real-valued, mean-zero, σ -sub-Gaussian stochastic process. Let $\{X_i\}_{i=1}^\infty$ be an $\mathbb{R}^d$ -valued stochastic process.

Define $\Sigma_t = \Sigma_0 + \sum_{i=1}^t X_i X_i^\top$. With probability at least $1 - \delta$, we have for all $t \geq 1$:

$$\left\| \sum_{i=1}^t X_i \varepsilon_i \right\|_{\Sigma_t^{-1}}^2 \leq \sigma^2 \log \left(\frac{\det(\Sigma_t) \det(\Sigma_0)^{-1}}{\delta^2} \right).$$

Completing the Confidence Bound

For all t , with probability at least $1 - \delta$, we plug the lemma back into our triangle inequality bound:

$$\begin{aligned}\|\hat{\mu}_t - \mu^*\|_{\Sigma_t} &\leq \sqrt{\lambda} \|\mu^*\|_2 + \left\| \sum_{\tau=0}^{t-1} \eta_\tau x_\tau \right\|_{\Sigma_t^{-1}} \\ &\leq \sqrt{\lambda} \|\mu^*\|_2 + \sqrt{\sigma^2 \log \left(\frac{\det(\Sigma_t)}{\det(\Sigma_0) \delta^2} \right)}.\end{aligned}$$

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Substitute our log-determinant bound (Lemma 7) into the term above:

$$\log \left(\frac{\det(\Sigma_t)}{\det(\Sigma_0)} \right) \leq d \log \left(1 + \frac{tB^2}{d\lambda} \right)$$

This matches our definition of $\sqrt{\beta_t}$, concluding that $\mu^* \in \text{BALL}_t$.

Beyond Linear: Neural Network Reward Models

What if our expected reward is a non-linear neural network $f(x; \theta)$?

$$\mathbb{E}[r_t \mid x_t = x] = f(x; \theta^*)$$

Optimizing UCB exploration directly in deep neural networks is notoriously hard. But in the **overparameterized regime**, neural networks behave like linear models!

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The Local Linear Approximation: Consider a highly overparameterized network initialized at θ_0 . We take a first-order Taylor expansion around the initialization:

$$f(x; \theta) \approx f(x; \theta_0) + \langle \nabla_{\theta} f(x; \theta_0), \theta - \theta_0 \rangle$$

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This allows us to extract a **fixed feature map** driven by the network architecture:

$$\phi(x) := \nabla_{\theta} f(x; \theta_0) \in \mathbb{R}^p$$

where p is the number of parameters.

Neural UCB & The Neural Tangent Kernel

Using the gradient feature map $\phi(x) = \nabla_{\theta} f(x; \theta_0)$, we can define a kernel:

$$k_{\text{NTK}}(x, x') := \phi(x)^{\top} \phi(x') = \langle \nabla_{\theta} f(x; \theta_0), \nabla_{\theta} f(x'; \theta_0) \rangle$$

This is the **Neural Tangent Kernel (NTK)**.

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Neural UCB (NTK Bandits): We can simply run LinUCB (or GP-UCB) using these extracted features!

$$\Lambda_t = \lambda I + \sum_{\tau=0}^{t-1} \phi(x_{\tau}) \phi(x_{\tau})^{\top}$$

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$$\Lambda_t = \lambda I + \sum_{\tau=0}^{t-1} \phi(x_{\tau}) \phi(x_{\tau})^{\top}$$
$$b_t(x) = \beta \sqrt{\phi(x)^{\top} \Lambda_t^{-1} \phi(x)} \quad (\text{or use the kernel trick})$$

Neural UCB & The Neural Tangent Kernel

Using the gradient feature map $\phi(x) = \nabla_{\theta} f(x; \theta_0)$, we can define a kernel:

$$k_{\text{NTK}}(x, x') := \phi(x)^{\top} \phi(x') = \langle \nabla_{\theta} f(x; \theta_0), \nabla_{\theta} f(x'; \theta_0) \rangle$$

This is the **Neural Tangent Kernel (NTK)**.

Neural UCB (NTK Bandits): We can simply run LinUCB (or GP-UCB) using these extracted features!

$$\Lambda_t = \lambda I + \sum_{\tau=0}^{t-1} \phi(x_{\tau}) \phi(x_{\tau})^{\top}$$
$$b_t(x) = \beta \sqrt{\phi(x)^{\top} \Lambda_t^{-1} \phi(x)} \quad (\text{or use the kernel trick})$$
$$x_t = \operatorname{argmax}_{x \in D} \left(f(x; \theta_{t-1}) + b_t(x) \right)$$

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Takeaway: In the infinite-width limit, training a neural network with gradient descent is equivalent to kernel regression with the NTK. By using NTK features for our bonuses, we achieve theoretically